

Challenge (Habanero)

Q1. If $2^x = 32$, what is x ?

- (A) 3
- (B) 4
- (C) 5
- (D) 6
- (E) 7

Q2. Solve for y : $5^y = 125$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q3. A recipe for making cookies requires 3^x cups of sugar. If $3^x = 27$, how many cups of sugar are needed?

- (A) 3
- (B) 6
- (C) 9
- (D) 12
- (E) 15

Q4. A building is designed with a height of 2^{x+1} meters. If the height is 16 meters, what is x ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q5. If $x^2 = 4^x$, what are the possible values of x ?

- (A) 0, 1
- (B) 1, 2
- (C) 2, 4
- (D) 1, 4
- (E) 0, 2

Q6. Solve for z : $2^{z+2} = 32$

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q7. A water tank is filled at a rate of 3^{x-1} liters per minute. If 9 liters are filled in one minute, what is x ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q8. If $4^x = 2^{2x}$, what is x ?

- (A) 0
- (B) 1
- (C) 2
- (D) 4
- (E) 8

Open Ended Questions (FRQ)

Q9. A chef needs to package 5^x cookies into boxes that hold 25 cookies each. If she has 125 cookies, what is x ? Show your work.

Q10. An architect designs a building with 2^x floors, and each floor has 3^y rooms. If the building has 12 floors and 27 rooms per floor, what are the values of x and y ? Show your work.

Chef's Tips

- Ensure students show their work for FRQs
- Emphasize the importance of checking units in word problems
- Encourage students to use different methods to solve exponential equations